The empirical recurrence for OEIS A232157, recovered from its tail

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Abstract

OEIS A232157 records "Empirical recurrence of order 94" and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 328$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 4$, so the recurrence is proved for every $n \geq 98$. Its last coefficient is $c_{94} = -1 \neq 0$, so no further tail satisfies anything shorter; any order-94 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry's recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A232157 is "Number of nX4 0..3 arrays with every 0 next to a 1, every 1 next to a 2 and every 2 next to a 3 horizontally, vertically or antidiagonally, and no adjacent values equal.". It has offset 1 and begins

$$8, 218, 1284, 7770, 46702, 286770, 1767266, 10830650, \ldots$$

The entry states:

> Empirical recurrence of order 94 (see link above)

As of the "Last modified" line on the live entry (Jul 23 2025 06:55:58, revision 7) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry's condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u, and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) \;=\; \iota^{\top} M^{\,j(n)} \tau$$

for a linear reindexing j fixed by the entry's shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 328$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 328$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \ldots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S'+1$ consecutive values of a. $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 94: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 4$ is the first whose minimal order is exactly the stated 94.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 656$ exact terms from $n = 4$ on, it returns order 94 — the order the entry states — with integer coefficients:

$$a(n) \;=\; \sum_{i=1}^{94} c_i\, a(n-i),$$

c_1	16	c_{25}	−4579175244	c_{49}	−801725870746	c_{73}	9728140277
c_2	−88	c_{26}	17753676344	c_{50}	499289397746	c_{74}	−827781365
c_3	155	c_{27}	−7893511689	c_{51}	828379788467	c_{75}	−6195062463
c_4	92	c_{28}	−23264349930	c_{52}	−382679611244	c_{76}	−5713700538
c_5	1282	c_{29}	29818570032	c_{53}	−833298022756	c_{77}	−1751717675
c_6	−12149	c_{30}	9064359628	c_{54}	237367914991	c_{78}	704101864
c_7	16402	c_{31}	−38761633302	c_{55}	789352840022	c_{79}	780156624
c_8	111466	c_{32}	18415196618	c_{56}	−140455939797	c_{80}	268201567
c_9	−467296	c_{33}	9683305551	c_{57}	−765102409949	c_{81}	32481807
c_{10}	252101	c_{34}	−26523887047	c_{58}	−1238951153	c_{82}	−12100034
c_{11}	2659772	c_{35}	55695868193	c_{59}	713762825030	c_{83}	−13600343
c_{12}	−7608106	c_{36}	−14257706819	c_{60}	185999081217	c_{84}	−8220618
c_{13}	3169554	c_{37}	−130982757110	c_{61}	−530578706799	c_{85}	−1449739
c_{14}	26870941	c_{38}	96841830119	c_{62}	−228949111494	c_{86}	939716
c_{15}	−65152319	c_{39}	205295059142	c_{63}	362051816462	c_{87}	93466
c_{16}	25517050	c_{40}	−191460327559	c_{64}	192238390861	c_{88}	−198998
c_{17}	177405832	c_{41}	−294495608812	c_{65}	−253577825204	c_{89}	5099
c_{18}	−404652905	c_{42}	280662422894	c_{66}	−193027062217	c_{90}	21484
c_{19}	137899964	c_{43}	420982020109	c_{67}	95576919170	c_{91}	−2602
c_{20}	1020388060	c_{44}	−365306373251	c_{68}	114561002176	c_{92}	−973
c_{21}	−1994423166	c_{45}	−585115267441	c_{69}	−12391806695	c_{93}	195
c_{22}	22477446	c_{46}	442426116264	c_{70}	−32319970960	c_{94}	−1
c_{23}	5188381887	c_{47}	734262925791	c_{71}	11946521263		
c_{24}	−6358106185	c_{48}	−505517822262	c_{72}	22645009446		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 98$, and it is the recurrence OEIS A232157 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 4$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{94} - c_1 x^{94-1} - \cdots - c_{94}$. Its constant term is $-c_{94} = 1$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 94 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 94, so $\deg q = 94$, and it is monic. It holds from some index t on. From $\max(t, 4)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 94, so $q = q_{\min}$. □

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 19 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry's English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 94, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry's.

Third, the recovered recurrence was evaluated directly on the entry's own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is -1.

Remark 4. The conjecture's own statement was never read. The entry pins it down to the family of order-94 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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